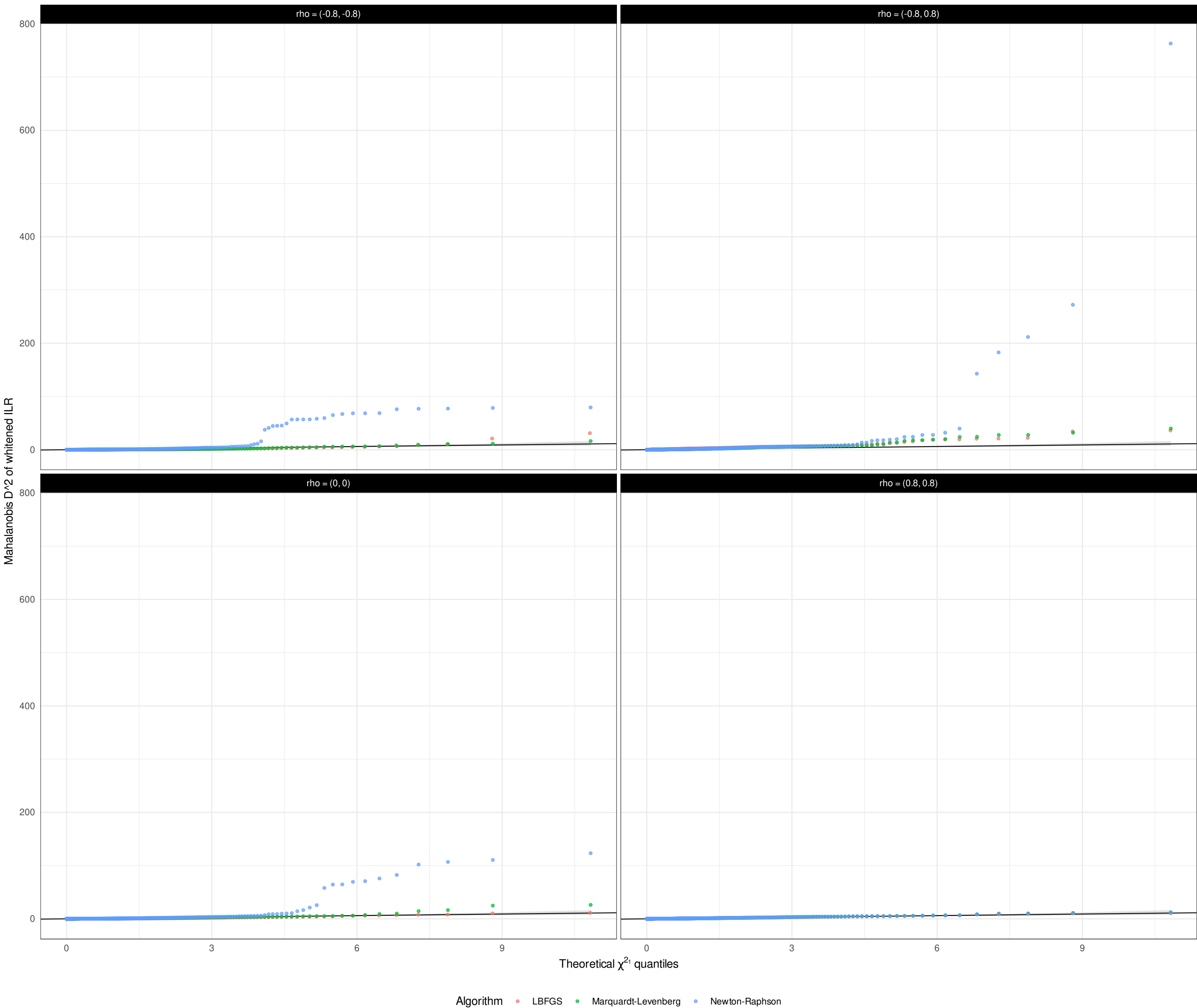
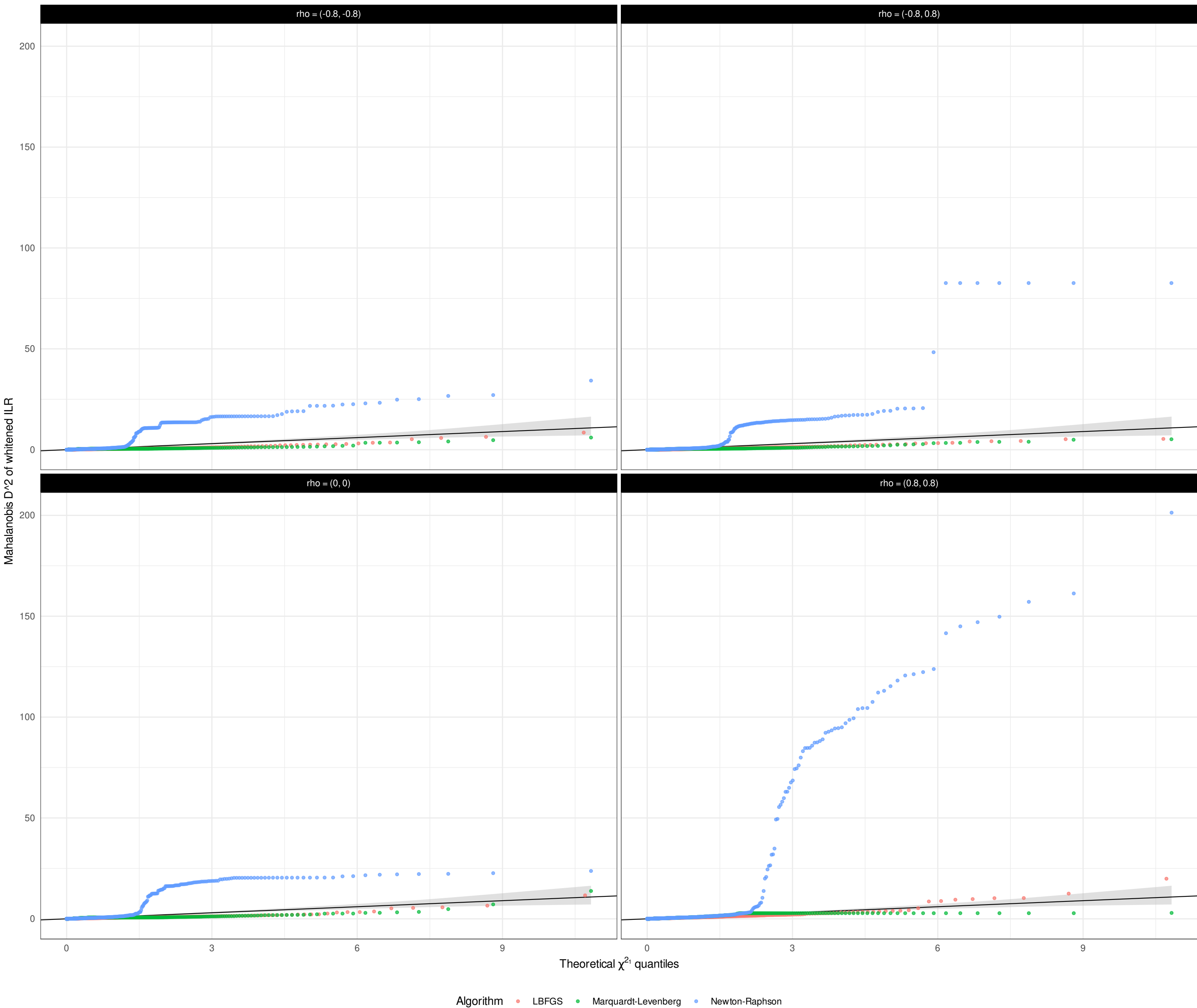
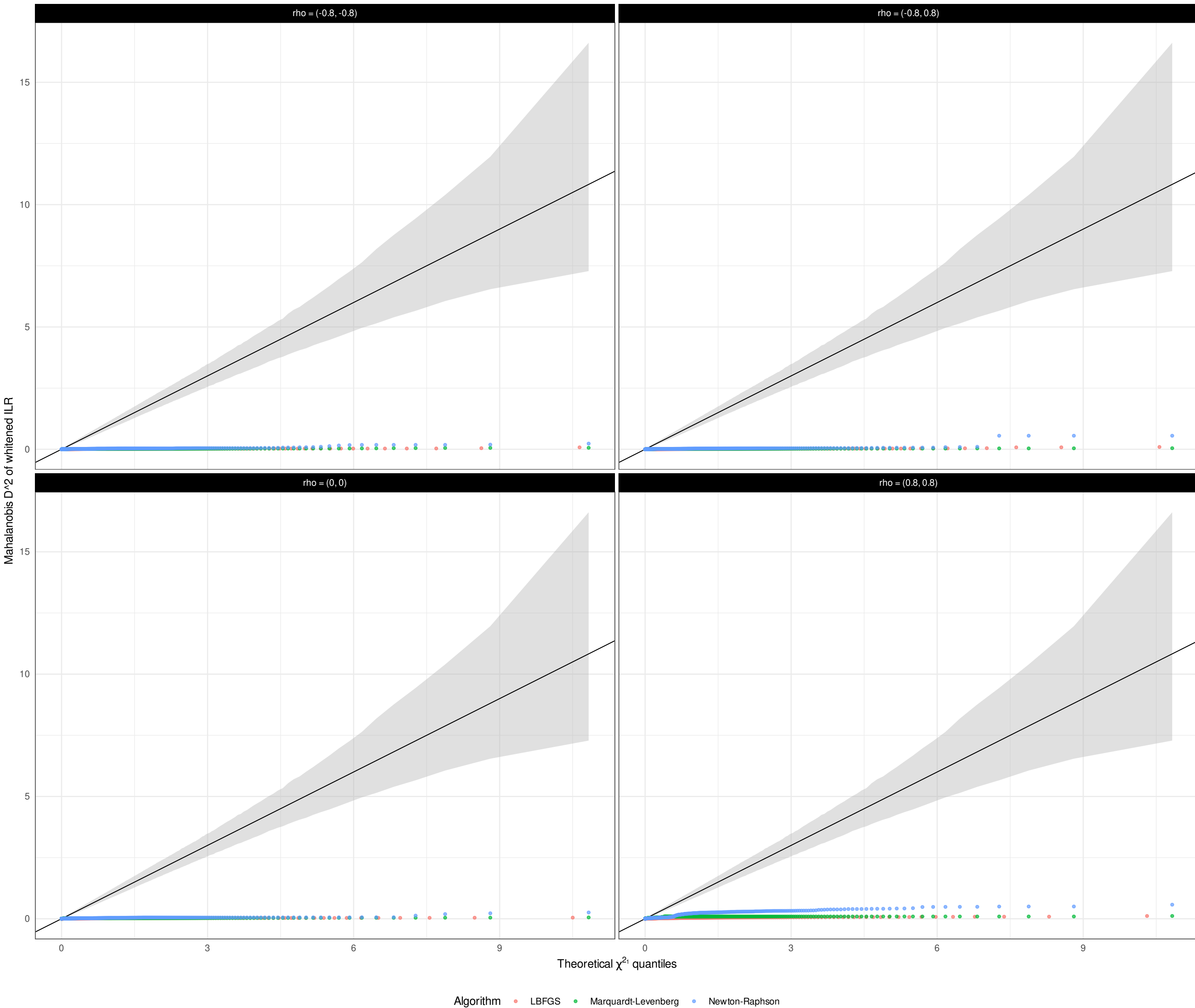
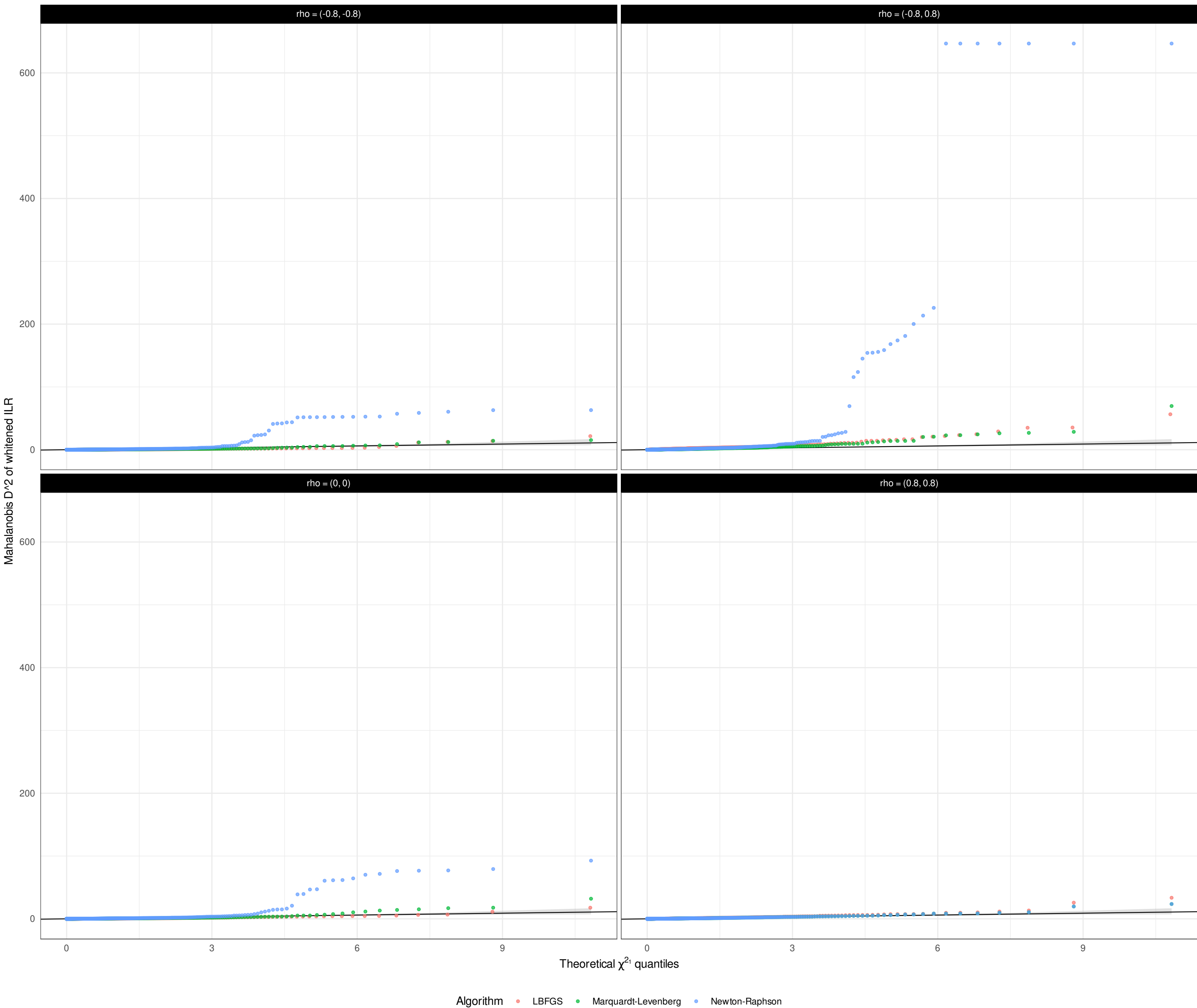
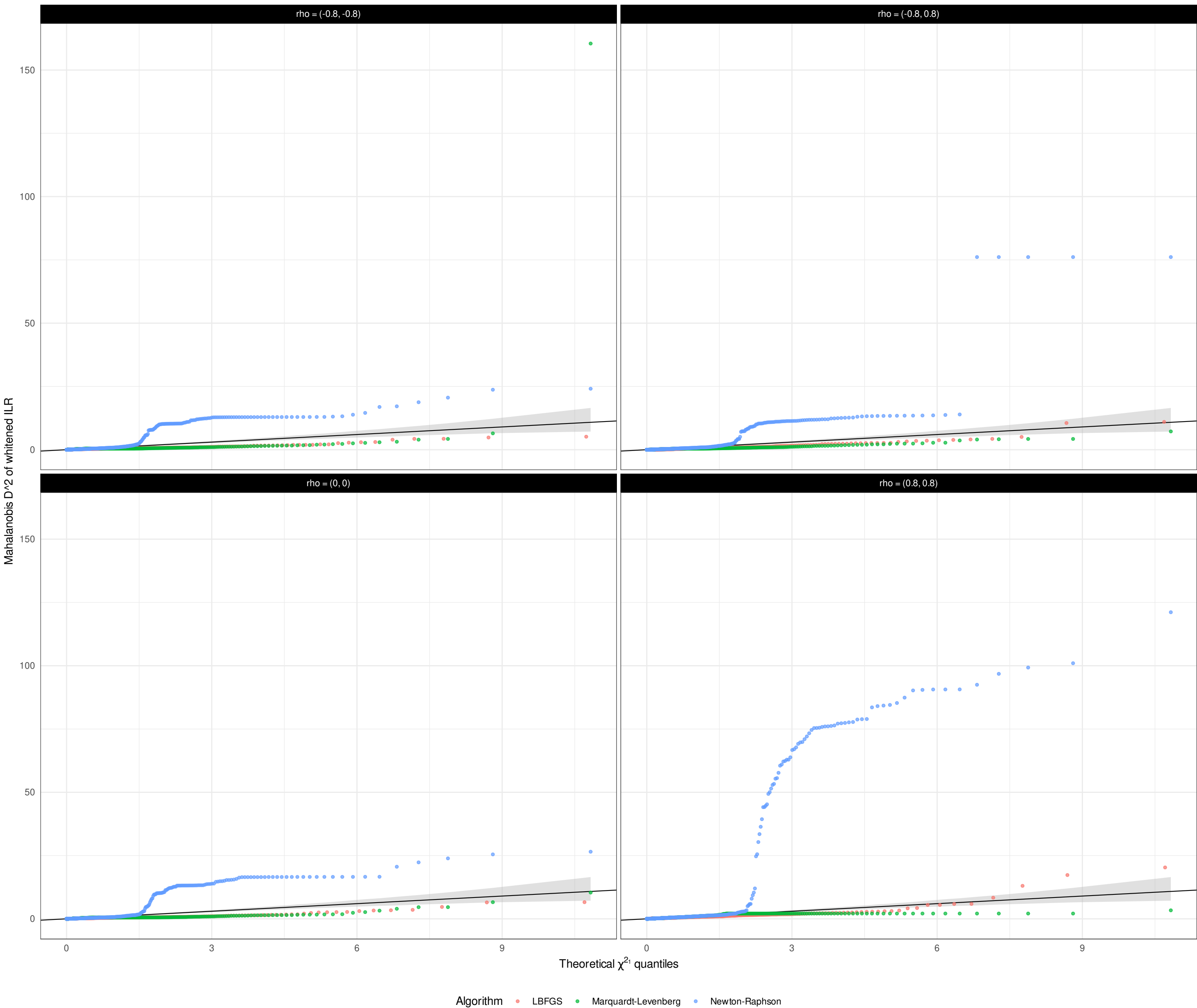




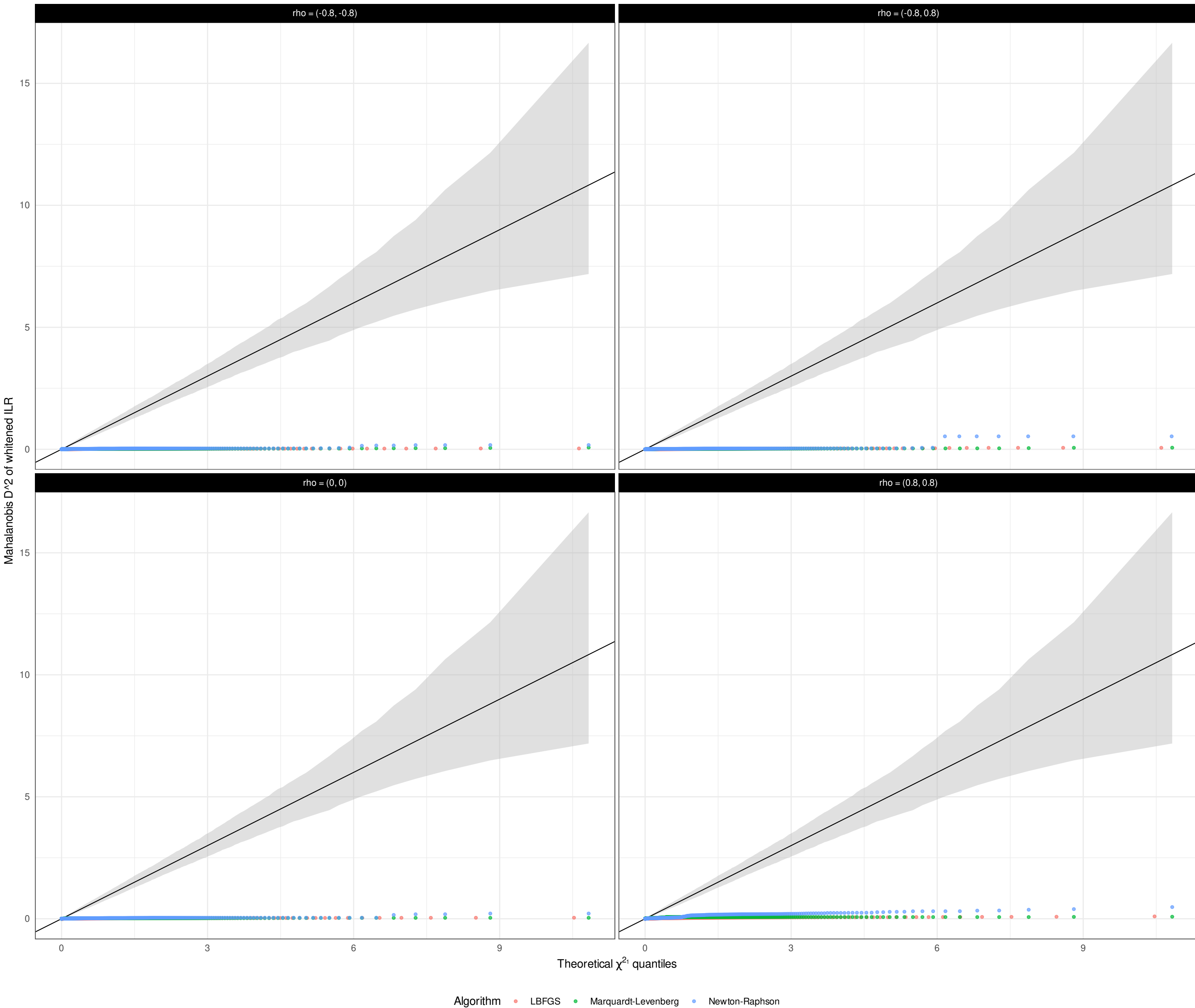




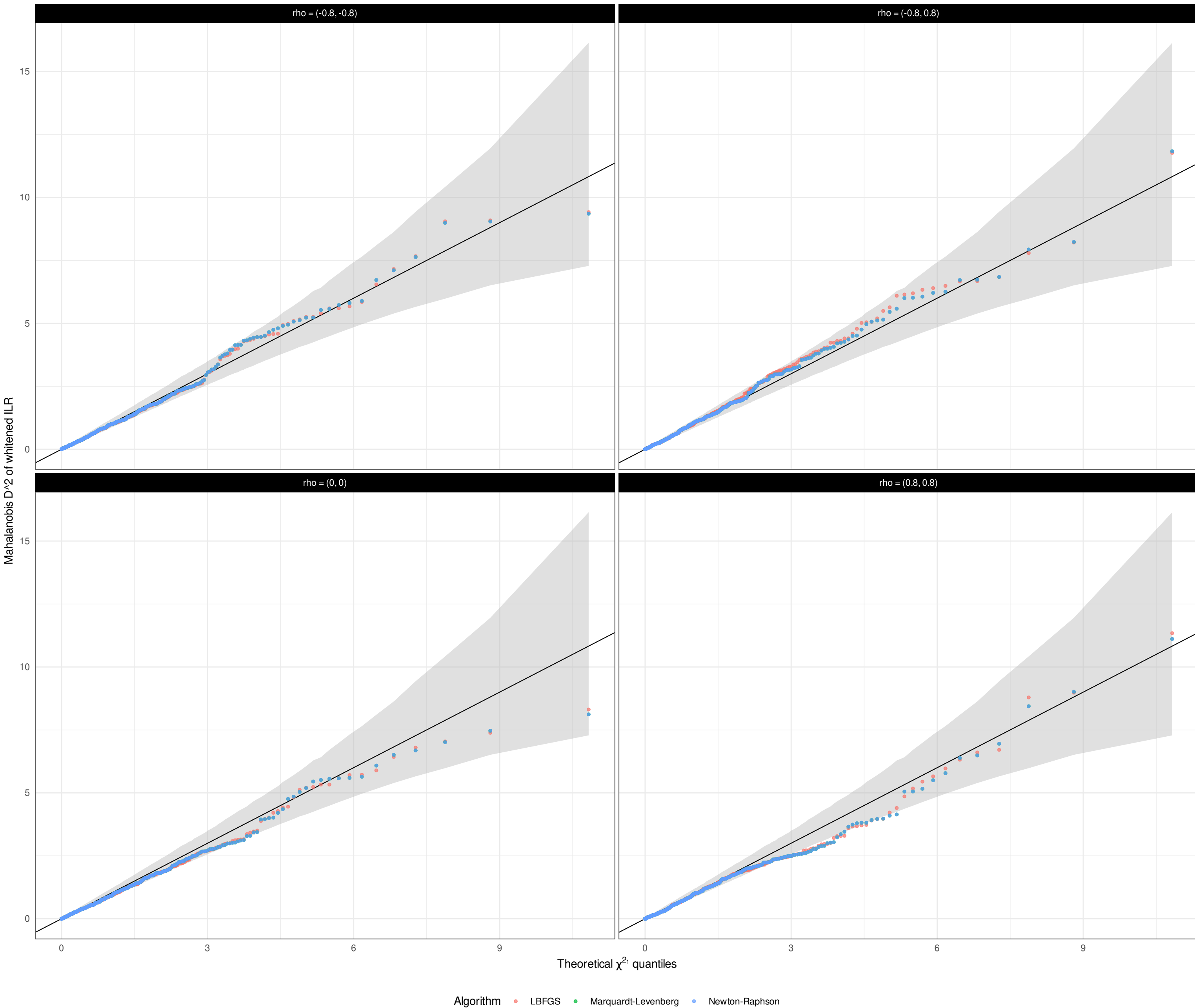


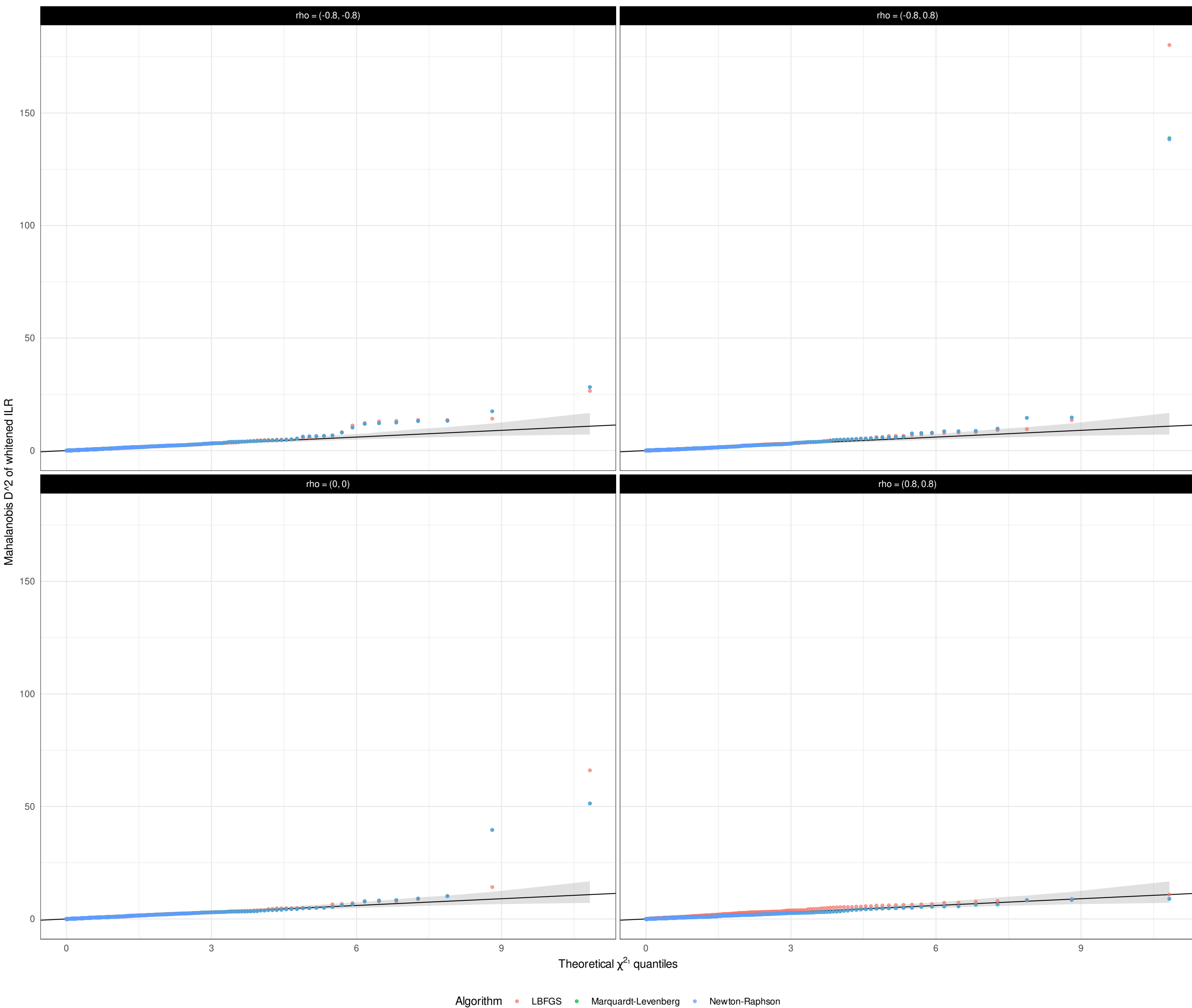


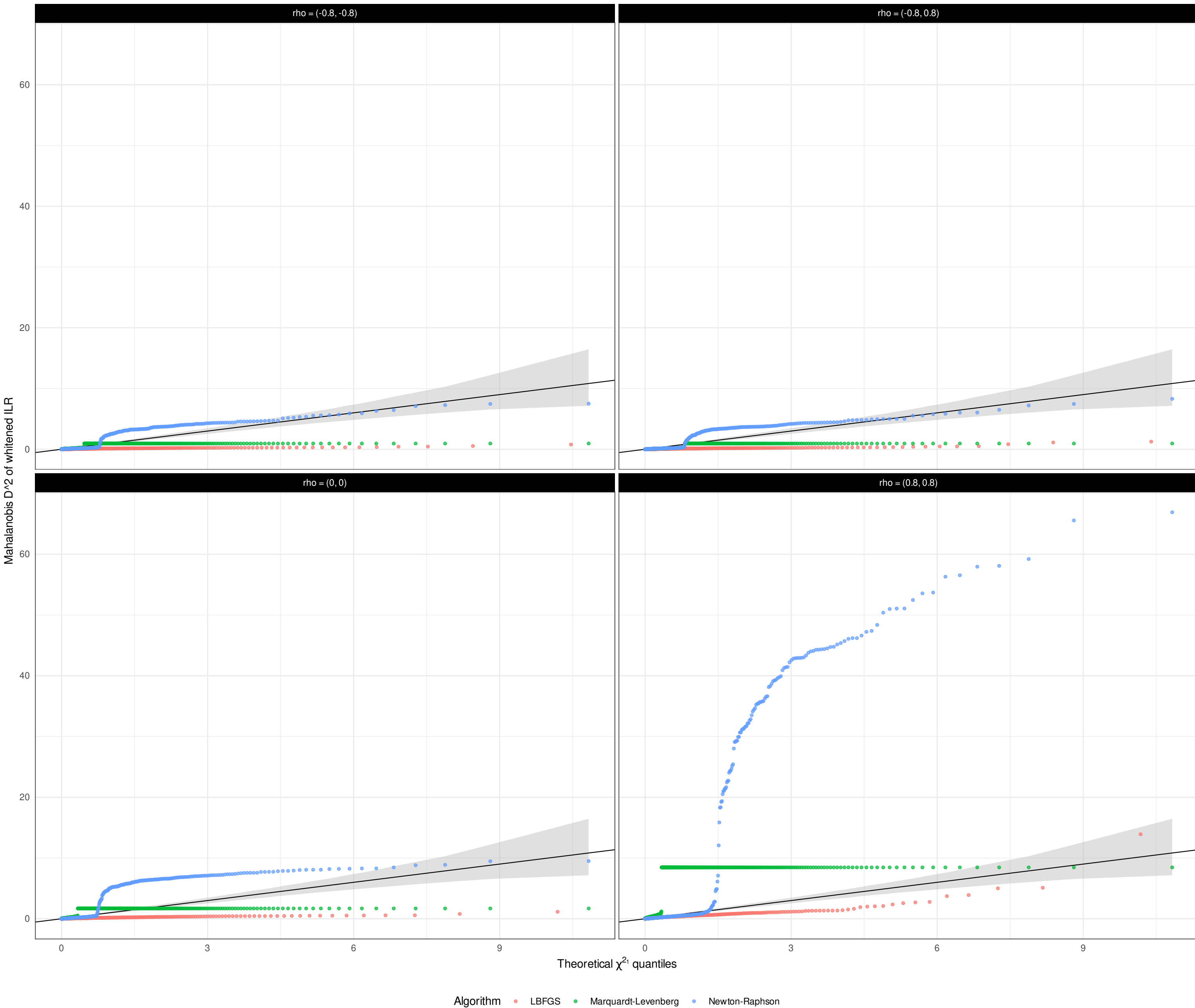
small_CLD / heteroscedastic / highly unbalanced

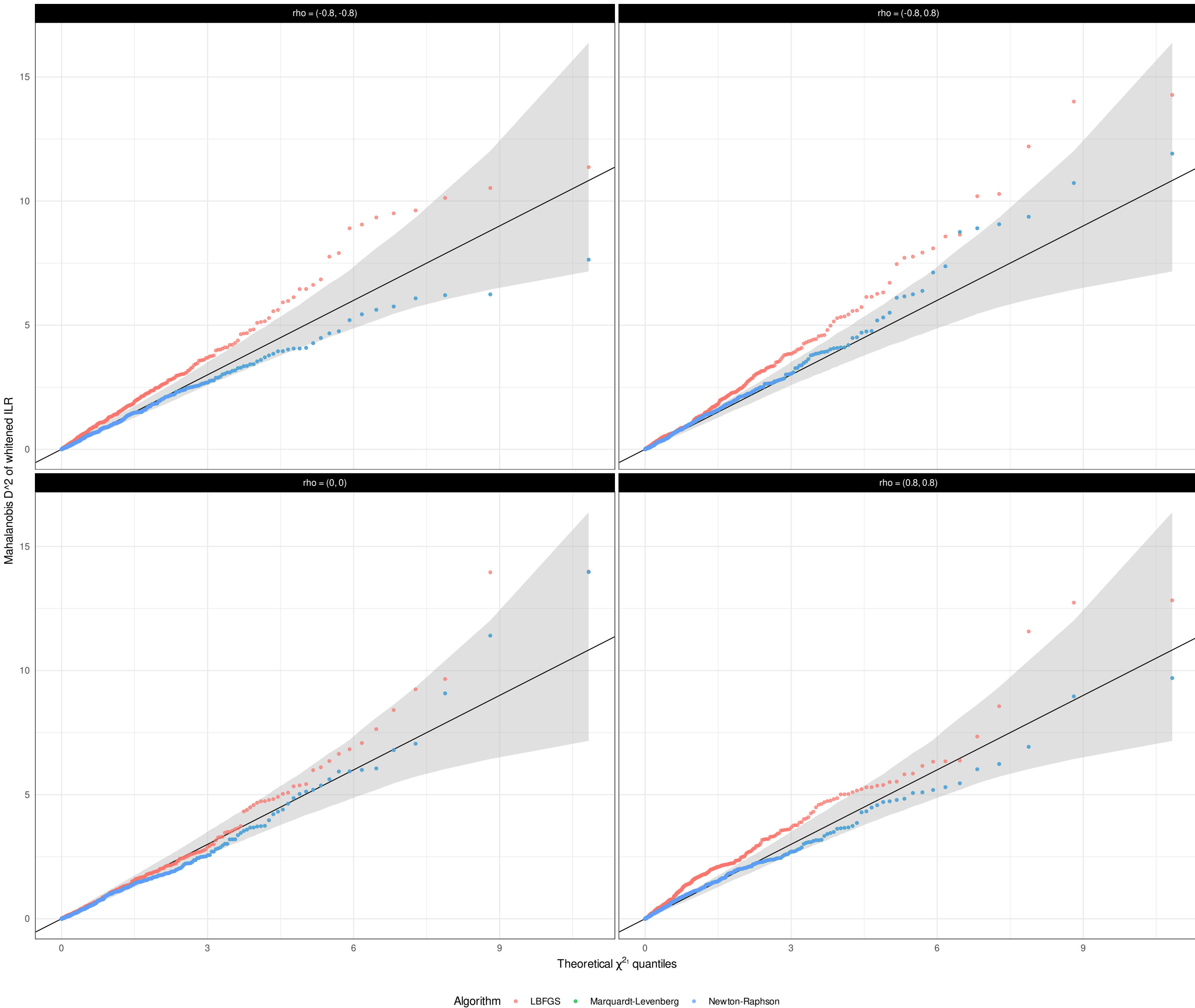


$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.

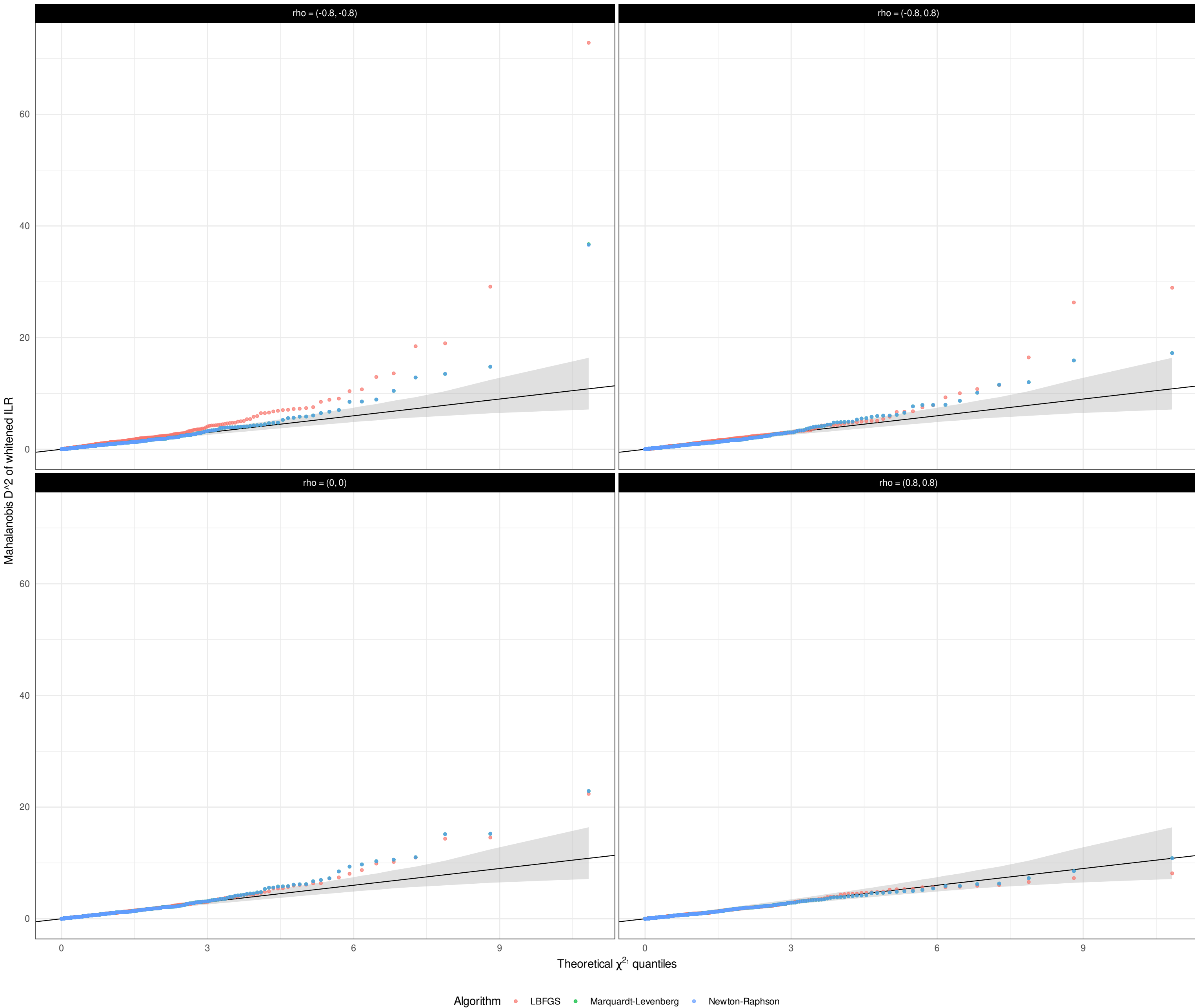






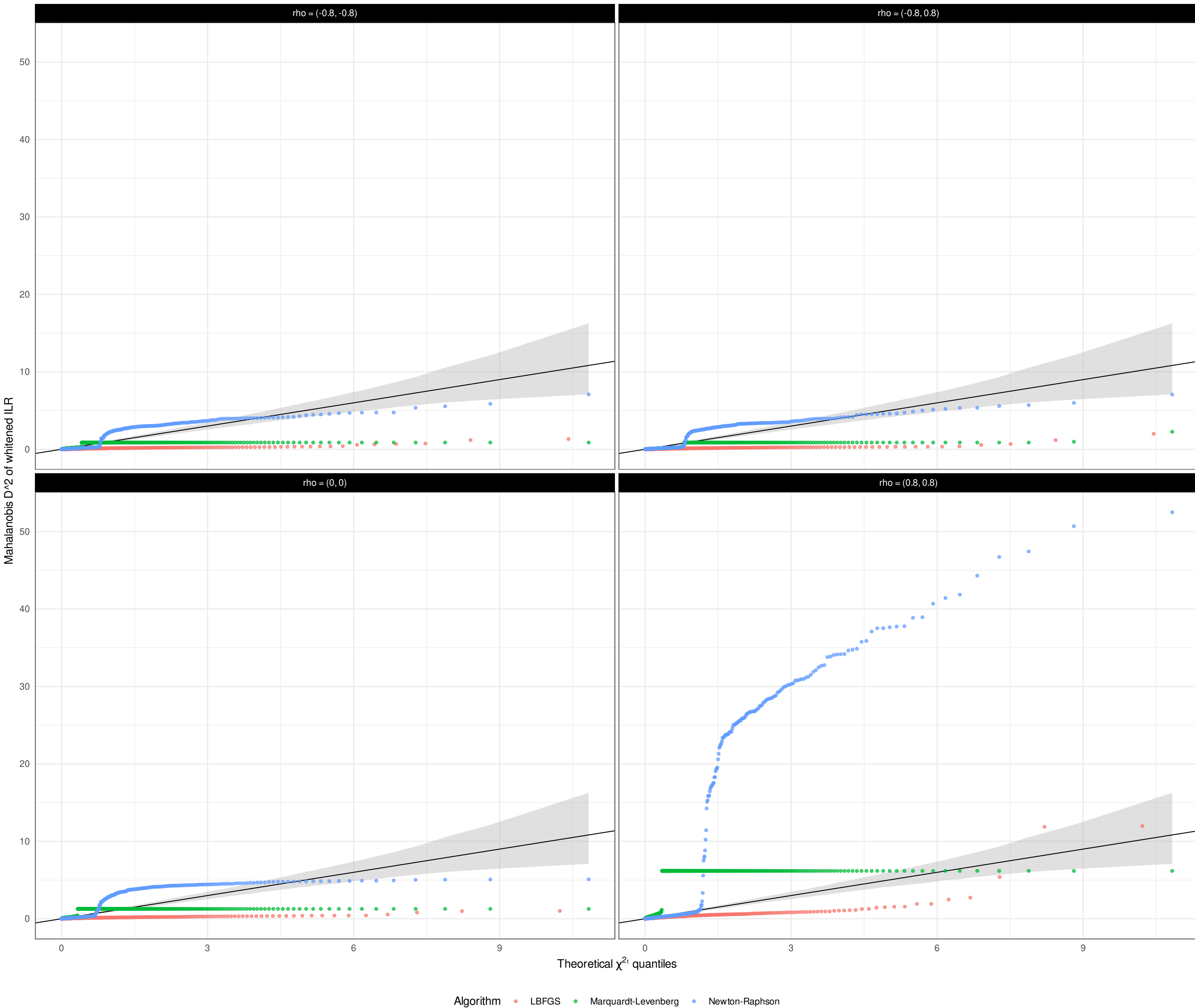


large_CLD / heteroscedastic / moderately unbalanced



$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.

large_CLD / heteroscedastic / highly unbalanced



$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.